

CPU performance analysis for MySQL using Hot/Cold Flame Graphs

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About Percona

- Founded in 2006
- 190+ staff worldwide
- Percona blog has **100k+ readers** per month
- Percona tools have been used **over a billion times** in the past 3 years
- Percona conferences attract **thousands of attendees** from some of the world's largest companies (**#PerconaLive**)
- Percona software is installed on **millions of servers**

About me

- Support Engineer at Percona since 2017.
- Working with MySQL for over six years.
- Working with databases for over nine years.
- Speaker at global conferences and meetups about MySQL/MongoDB.
- Social media
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So Brendan

Or "What are FlameGraphs?"

What are FlameGraphs

Flame graphs are a visualization of profiled software, allowing the most frequent code-paths to be identified quickly and accurately

~ Brendan Gregg

<http://www.brendangregg.com/FlameGraphs/cpuflamegraphs.html>

What are FlameGraphs: code path

```
int main() {  
    ...  
    GetBookList(book);  
    ...  
}  
  
void GetBookList(Book book) {  
    ...  
    while (book =  
        GetBook(book);  
        ...  
}
```

```
void GetBook(Book book) {  
    a = GetAuthor(book);  
    pub = GetPublisher(book);  
    p =  
    GetPrints(book).toString();  
  
    printf("%s published by  
    %s [prints: %s]", a, pub, p);  
    ...  
}
```

What are FlameGraphs: code path

Possible call chains

main() -> GetBooksList() -> GetBook() -> GetAuthor()...

main() -> GetBooksList() -> GetBook() -> GetEditorial()...

main() -> GetBooksList() -> GetBook() -> GetPrints()...

What are FlameGraphs: profiling

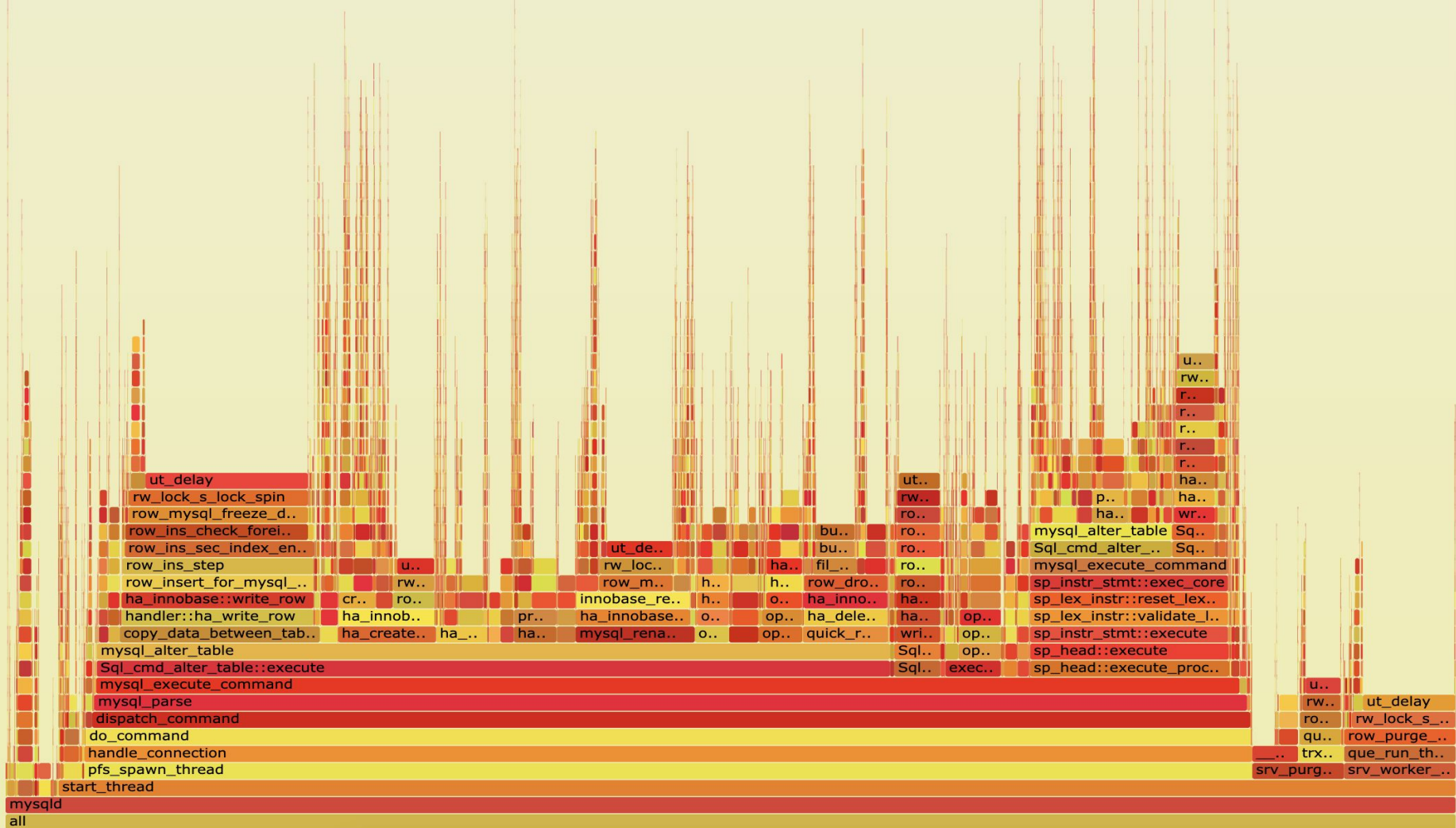
Captures of metrics of some type for later aggregation.

What are FlameGraphs

```
# profile it
perf record -a -g -F99 -p $(pgrep -x mysqld) --
sleep 60
```

```
# make it machine-readable
perf script > s.out
```

```
# graph it!
stackcollapse-perf.pl s.out | flamegraph.pl > s.svg
```

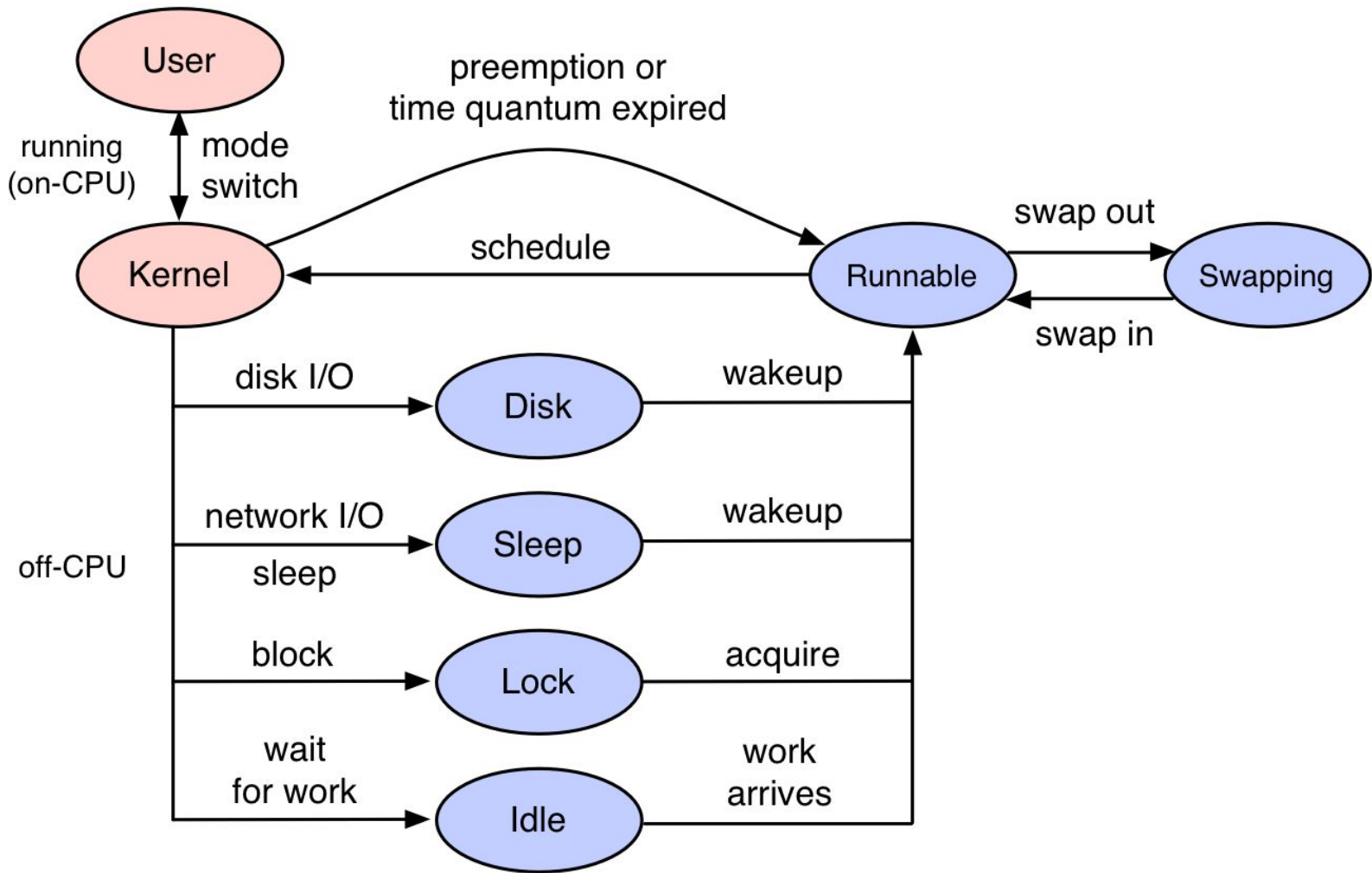


Cold Graphs

What are ColdGraphs / Off-CPU analysis

Off-CPU analysis is a performance methodology where off-CPU time is measured and studied, along with context such as stack traces.

<http://www.brendangregg.com/offcpuanalysis.html>



First things First

RHEL

```
yum install bcc-tools
```

Ubuntu

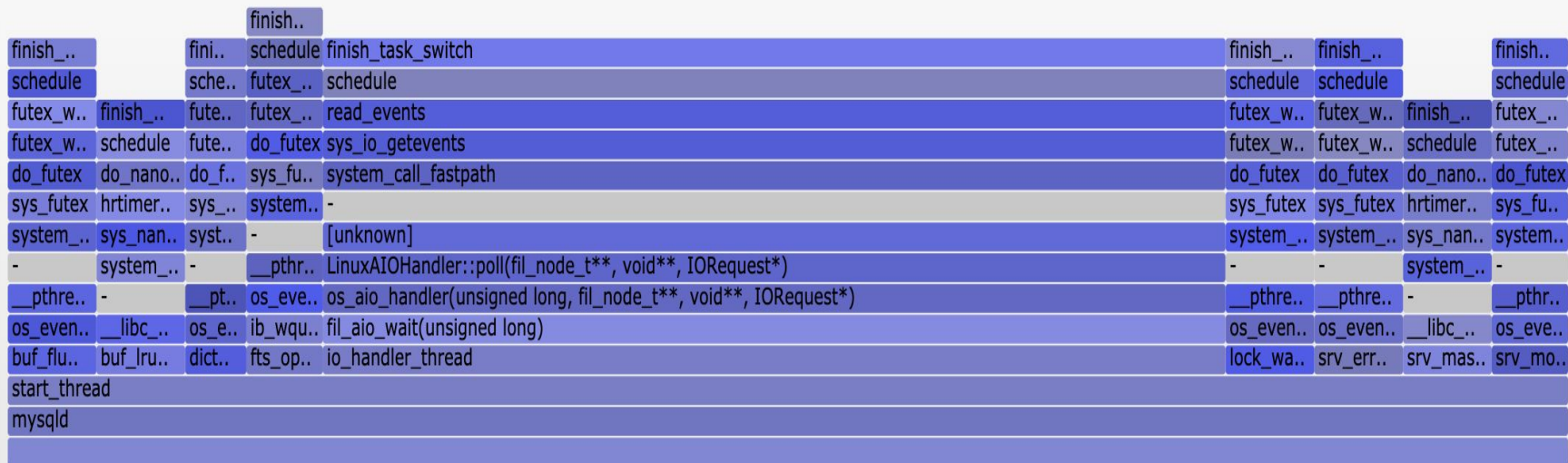
```
sudo apt-get install bpfcc-tools linux-headers-$(uname -r)
```

Validating

```
ls -lart /usr/share/bcc/tools/
```

Profiling

```
# /usr/share/bcc/tools/offcputime -df -p `pgrep -nx mysqld` 30 > out.stacks  
# git clone https://github.com/brendangregg/FlameGraph  
# cd FlameGraph  
# ./flamegraph.pl --color=io --title="Off-CPU Time Flame Graph" --countname=us <  
out.stacks > out.svg
```





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Thank you

Grazie

Obrigado

Gracias

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